3D-PLANES

SYNOPSIS

Equation of a Plane:

- Every first degree equation in x,y,z always represents a plane.
- Plane surface is a surface in which line joining every two points P and Q on it lies entirely in the surface.
- The general form of equation of plane is $ax+by+cz+d=0$, a,b,c are not all zero i.e., $a^2+b^2+c^2 \neq 0$

Equation of Planes with Different Conditions:

- i) The equation of the plane passing through the point (x_1,y_1,z_1) and having d.r's of normal as (a,b,c) is $a(x-x_1)+b(y-y_1)+c(z-z_1)=0$

  or $ax+by+cz=ax_1+by_1+cz_1$

  ii) The equation of the plane passing through a point (x_1,y_1,z_1) and parallel to the plane $ax+by+cz+d=0$ is

  $a(x-x_1)+b(y-y_1)+c(z-z_1)=0$

  $\Rightarrow ax+by+cz=ax_1+by_1+cz_1$

W.E-1 : The equation of the plane parallel to the plane 2x+3y+4z+5=0 and passing through the point (1,1,1) is

Sol : The plane is

$$a(x-x_1)+b(y-y_1)+c(z-z_1)=0$$

$$2(x-1)+3(y-1)+4(z-1)=0$$

$$\Rightarrow 2x+3y+4z-9=0$$

Equation of plane which is Parallel to lines:

- i) The equation of the plane passing through the point (x_1,y_1,z_1) and parallel to lines whose d.r's are (a_1,b_1,c_1) and (a_2,b_2,c_2) is

$$\begin{vmatrix} x-x_1 & y-y_1 & z-z_1 \\ a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \end{vmatrix}=0$$

ii) The equation of the plane passing through the points (x_1,y_1,z_1), (x_2,y_2,z_2) and parallel to the line whose d.r's are (a,b,c) is

$$\begin{vmatrix} x-x_1 & y-y_1 & z-z_1 \\ x_2-x_1 & y_2-y_1 & z_2-z_1 \\ a & b & c \end{vmatrix}=0$$

iii) The equation of the plane passing through three non collinear points $(x_1,y_1,z_1),(x_2,y_2,z_2),(x_3,y_3,z_3)$ is

$$\begin{vmatrix} x-x_1 & y-y_1 & z-z_1 \\ x_2-x_1 & y_2-y_1 & z_2-z_1 \\ x_3-x_1 & y_3-y_1 & z_3-z_1 \end{vmatrix}=0$$

iv) If $(x_1,y_1,z_1),(x_2,y_2,z_2)(x_3,y_3,z_3)$ and (x_4,y_4,z_4) are coplanar, then

$$\begin{vmatrix} x_4-x_1 & y_4-y_1 & z_4-z_1 \\ x_2-x_1 & y_2-y_1 & z_2-z_1 \\ x_3-x_1 & y_3-y_1 & z_3-z_1 \end{vmatrix}=0$$

General equation of a plane with different conditions:

- i) The equation of a plane with **d.r's** of normal as (a , b ,c) is $ax+by+cz+d=0$.

  ii) If a) a=0, $b\neq 0, c\neq 0$ Then equation $by+cz+d=0$ represents a plane which is parallel to x-axis and $\perp^{er}$ to YZ - plane.

  b) $b=0, a\neq 0, c\neq 0$ then equation $ax+cz+d=0$ represents a plane which is parallel to y-axis and $\perp^{er}$ to xz -plane.

  c) $a\neq 0, b\neq 0, c=0$ then equation $ax+by+d=0$ represents a plane which is parallel to z-axis and $\perp^{er}$ to XY -plane.

iii) The equation of the plane passing through (x_1, y_1, z_1) and parallel to

a) yz- plane and $\perp^{er}$ to X-axis is $x = x_1$

b) xy-plane and $\perp^{er}$ to Z-axis is $z = z_1$

c) zx-plane and $\perp^{er}$ to Y-axis is $y = y_1$

iv) Equation of plane parallel to the plane $ax + by + cz + d_1 = 0$ is of the form $ax + by + cz + d_2 = 0$

v) Distance between the above two parallel planes is $\dfrac{|d_1 - d_2|}{\sqrt{a^2 + b^2 + c^2}}$

vi) Equation of plane parallel to $\vec{r}.\vec{n} = d_1$ is $\vec{r}.\vec{n} = d_2$ (vector form)

vii) The equation of the plane, mid way between the parallel planes $ax + by + cz + d_1 = 0$ and $ax + by + cz + d_2 = 0$ is

$$ax + by + cz + \left(\frac{d_1 + d_2}{2}\right) = 0$$

viii) The equation of the plane which bisects the line joining $A(x_1, y_1, z_1)$ and $B(x_2, y_2, z_2)$ and perpendicular to AB is

$$(x_1 - x_2)x + (y_1 - y_2)y + (z_1 - z_2)z = \frac{(x_1^2 + y_1^2 + z_1^2) - (x_2^2 + y_2^2 + z_2^2)}{2}$$

ix) The reflection of $a^1x + b^1y + c^1z + d^1 = 0$ in the plane $ax + by + cz + d = 0$ is given by

$$2(aa^1 + bb^1 + cc^1)(ax + by + cz + d) = (a^2 + b^2 + c^2)(a^1x + b^1y + c^1z + d^1)$$

W.E-2 : Distance between parallel planes $2x - 2y + z + 3 = 0$ ***and*** $4x - 4y + 2z + 5 = 0$ ***is***

Sol : The given planes are $2x - 2y + z + 3 = 0$,

$$2x - 2y + z + \frac{5}{2} = 0$$

Here $a = 2, b = -2, c = 1, d_1 = 3, d_2 = \frac{5}{2}$

$$\text{Distance} = \left|\frac{d_1 - d_2}{\sqrt{a^2 + b^2 + c^2}}\right|, = \frac{\left|3 - \frac{5}{2}\right|}{\sqrt{4 + 4 + 1}} = \frac{1}{6}$$

W.E-3 : The equation of the parallel plane lying midway between the parallel planes $2x - 3y + 6z - 7 = 0$ ***and*** $2x - 3y + 6z + 7 = 0$ ***is***

Sol : The required plane is $ax + by + cz + \dfrac{d_1 + d_2}{2} = 0$

$$2x - 3y + 6z + \frac{(-7 + 7)}{2} = 0$$

$$2x - 3y + 6z = 0$$

W.E-4 : The reflection of the plane in the plane $x - y + z - 3 = 0$ ***is***

Sol : The given planes are ,

$2x - 3y + 4z - 3 = 0$ $(a^1x + b^1y + c^1z + d^1 = 0)$

$x - y + z - 3 = 0$ $(ax + by + cz + d = 0)$

$\therefore$ Equation of the required plane be obtained using the fact reflection of $a^1x + b^1y + c^1z + d^1 = 0$ in the plane $ax + by + cz + d = 0$ is given by

$$2(aa^1 + bb^1 + cc^1)(ax + by + cz + d) = (a^2 + b^2 + c^2)(a^1x + b^1y + c^1z + d^1)$$

$\therefore$ The reflection is $4x - 3y + 2z - 15 = 0$

Foot and image:

- i) The foot of the perpendicular of the point $P(x_1, y_1, z_1)$ on the plane $ax + by + cz + d = 0$ is $Q(h, k, l)$ then

$$\frac{h - x_1}{a} = \frac{k - y_1}{b} = \frac{l - z_1}{c} = \frac{-(ax_1 + by_1 + cz_1 + d)}{a^2 + b^2 + c^2}$$

ii) If Q (h, k, l) is the image of the point $p(x_1, y_1, z_1)$ w.r.to the plane $ax + by + cz + d = 0$ then

$$\frac{h - x_1}{a} = \frac{k - y_1}{b} = \frac{l - z_1}{c} = \frac{-2(ax_1 + by_1 + cz_1 + d)}{a^2 + b^2 + c^2}$$

iii) If 'd' is the distance from the origin and (l,m,n) are the dc's of the normal to the plane through the origin, then the foot of the perpendicular is (ld,md,nd)

W.E-5 : The foot of the perpendicular from the point P(1,3,4) to the plane $2x-y+z+3=0$ is

Sol : The given plane is $2x-y+z+3=0$

Here $a=2, b=-1, c=1, d=3$,

$(x_1,y_1,z_1)=(1,3,4)$

$$\therefore \frac{h-x_1}{a}=\frac{k-y_1}{b}=\frac{l-z_1}{c}=\frac{-(ax_1+by_1+cz_1+d)}{a^2+b^2+c^2}$$

$\therefore (h,k,l)=(-1,4,3)$

W.E-6 : If the image of the point (-1,3,4) in the plane $x-2y=0$ is (x_1,y_1,z_1) then $z_1=$

Sol : Given that $(h,k,l)=(x_1,y_1,z_1)$

$(x_1,y_1,z_1)=(-1,3,4)$

$$\therefore \frac{x_1+1}{1}=\frac{y_1-3}{-2}=\frac{z_1-4}{0}=-\left\{\frac{(-1)-(2)(3)}{1^2+(-2)^2+0^2}\right\}$$

$\Rightarrow z_1-4=0, \Rightarrow z_1=4$

Ratio formula:

➢ i) The ratio in which the plane ax + by + cz + d = 0 divides the line segment joining (x_1,y_1,z_1) and (x_2,y_2,z_2) is

$-(ax_1+by_1+cz_1+d):(ax_2+by_2+cz_2+d)$

ii) Position of the points w.r.to the plane

a) If $\frac{ax_1+by_1+cz_1+d}{ax_2+by_2+cz_2+d}>0$ then the points $P(x_1,y_1,z_1)$ and $Q(x_2,y_2,z_2)$ lie on same side of the plane $ax+by+cz+d=0$

b) If $\frac{ax_1+by_1+cz_1+d}{ax_2+by_2+cz_2+d}<0$ then the points $P(x_1,y_1,z_1)$ and $Q(x_2,y_2,z_2)$ lie on opposite sides of the plane $ax+by+cz+d=0$

W.E-7 : If the plane $2x-3y+5z-2=0$ divides the line segment joining (1, 2, 3) and (2, 1, k) in the ratio 9 : 11 then k =

Sol : $\frac{-(ax_1+by_1+cz_1+d)}{ax_2+by_2+cz_2+d}=\frac{9}{11}$

$$-\frac{(2\times1-3\times2+5\times3-2)}{(2\times2-3\times1+5\times k-2)}=\frac{9}{11}, \therefore k=-2$$

Normal form of a plane:

➢ i) If (l,m,n) are the direction cosines of normal to plane π and p is the $\perp^{er}$ distance from origin to the plane then the equation of plane is $lx+my+nz=p$

ii) The normal form of the plane representing by the equation $ax+by+cz+d=0$ is

a) If d < 0

$$\frac{a}{\sqrt{a^2+b^2+c^2}}x+\frac{b}{\sqrt{a^2+b^2+c^2}}y+\frac{c}{\sqrt{a^2+b^2+c^2}}z=\frac{-d}{\sqrt{a^2+b^2+c^2}}$$

b) If d > 0

$$\frac{-a}{\sqrt{a^2+b^2+c^2}}x+\frac{-b}{\sqrt{a^2+b^2+c^2}}y+\frac{-c}{\sqrt{a^2+b^2+c^2}}z=\frac{d}{\sqrt{a^2+b^2+c^2}}$$

W.E-8 : If the equation of the plane $2x-3y+6z=7$ in the normal form is $lx+my+nz=p$ then $l+p=$

Sol : Equation of the plane in the normal form is

$$\frac{2x}{\sqrt{4+9+36}}+\frac{-3y}{\sqrt{4+9+36}}+\frac{6z}{\sqrt{4+9+36}}=\frac{7}{\sqrt{4+9+36}}$$

$$\frac{2}{7}x-\frac{3}{7}y+\frac{6}{7}z=1, (lx+my+nz=p)$$

$$\Rightarrow l+p=\frac{2}{7}+1=\frac{9}{7}$$

Perpendicular distance from point to the plane:

➢ i) The perpendicular distance from (x_1, y_1, z_1) to the plane ax + by + cz + d = 0 is $\frac{|ax_1+by_1+cz_1+d|}{\sqrt{a^2+b^2+c^2}}$

ii) The perpendicular distance of the plane ax+by+cz+d=0 from the origin is $\frac{|d|}{\sqrt{a^2+b^2+c^2}}$.

W.E-9 : If the perpendicular distance from (1, 2, 4) to the plane $2x+2y-z+k=0$ is 3 then k - 4 =

Sol : $\left|\frac{2+4-4+k}{\sqrt{4+4+1}}\right|=3 \Rightarrow K=7$, $k-4=7-4=3$

Intercept form of a plane:

➢ i) If a plane cuts X-axis at $A(a,0,0)$, Y-axis at $B(0,b,0)$ and Z-axis at $C(0,0,c)$ then a,b,c are called X-intercept,Y-intercept, Z-intercept of the plane.

ii) The equation of the plane in intercept form is

$$\frac{x}{a}+\frac{y}{b}+\frac{z}{c}=1$$

iii) If $ax+by+cz+d=0$ is a plane if $a\neq 0, b\neq 0, c\neq 0$ then

X-intercept $=-\frac{d}{a}$

Y-intercept $=-\frac{d}{b}$, Z-intercept $=-\frac{d}{c}$

iv) The equation of the plane whose intercepts are K times the intercepts made by the plane $ax+by+cz+d=0$ on corresponding axes is $ax+by+cz+kd=0$.

Areas:

➢ i) Area of the triangle formed by the plane $\frac{x}{a}+\frac{y}{b}+\frac{z}{c}=1$ with

a) X – axis , Y –axis is $\frac{1}{2}|ab|$ Sq. units

b) Y– axis, Z– axis is $\frac{1}{2}|bc|$ Sq. units

c) Z– axis, X– axis is $\frac{1}{2}|ca|$ Sq. units

ii) If the plane $\frac{x}{a}+\frac{y}{b}+\frac{z}{c}=1$ meets the co-ordinate axes in the points A,B,C. then the area of the triangle ABC is

$$\frac{1}{2}\sqrt{(ab)^2+(bc)^2+(ca)^2}.$$

W.E-10 : The area of the triangle formed by the plane $2x+3y+6z+9=0$ with Y - axis, Z-axis is (in Sq.units)

Sol : The plane is $\frac{x}{\frac{-9}{2}}+\frac{y}{-3}+\frac{z}{\frac{-3}{2}}=1$

Here $a=\frac{-9}{2}, b=-3, c=\frac{-3}{2}$

The area of the triangle $=\frac{1}{2}|bc|$

$=\frac{1}{2}\left|(-3)\left(\frac{-3}{2}\right)\right| = \frac{9}{4}$ Sq.units

W.E-11 : The plane $\frac{x}{2}+\frac{y}{3}+\frac{z}{4}=1$ cuts the axes in A, B,C then the area of the $\Delta\, ABC$ is (squ)

Sol : $a=2, b=3, c=4$

$\therefore$ Area of the $\Delta ABC=$

$\frac{1}{2}\sqrt{(ab)^2+(bc)^2+(ca)^2}$, $=\sqrt{61}$.

Angle between Two Planes:

➢ i) The angle between two planes is equal to the angle between the perpendiculars from the origin to the planes.

ii) If 'θ' is the angle between the planes

$a_1x+b_1y+c_1z+d_1=0$ and
$a_2x+b_2y+c_2z+d_2=0$ then

$$\cos\theta=\pm\frac{a_1a_2+b_1b_2+c_1c_2}{\sqrt{a_1^2+b_1^2+c_1^2}\sqrt{a_2^2+b_2^2+c_2^2}}$$

iii) If the above two planes are parallel then

$$\frac{a_1}{a_2}=\frac{b_1}{b_2}=\frac{c_1}{c_2}$$

iv) If the above two planes are perpendicular then
$a_1a_2+b_1b_2+c_1c_2=0$

v) Angle between the line with d.c's (l_1,m_1,n_1) and the plane whose normal with d.c's (l_2,m_2,n_2) is θ then

$\cos(90-\theta)=|l_1l_2+m_1m_2+n_1n_2|$

vi) If θ is angle between a line L and a plane π then the angle between L and normal to the plane π is $90\pm\theta$.

W.E-12 : *If the angle between $x+2y+4z+7=0$ and $x-3y+2z+6=0$ is $Cos^{-1}\left(\sqrt{k}\right)$, then $k=$*

Sol : $$\cos\theta=\left|\frac{a_1a_2+b_1b_2+c_1c_2}{\sqrt{a_1^2+b_1^2+c_1^2}\sqrt{a_2^2+b_2^2+c_2^2}}\right|$$

Where $a_1=1;b_1=2;c_1=4$ **and**

$a_1=1;b_2=-3;c_2=2$

$$\therefore\cos\theta=\frac{|1\times1+(2)(-3)+(4)(2)|}{\sqrt{1^2+2^2+4^2}\sqrt{1^2+(-3)^2+(2)^2}}$$

$$=\frac{3}{\sqrt{21}\sqrt{14}}=\sqrt{\frac{9}{21\times14}}=\sqrt{\frac{3}{98}},\therefore k=\frac{3}{98}$$

W.E-13 : *If the planes $2x-y-3z-7=0$ and $4x-2y+5kz+9=0$ are parallel then $5k^2+6=$*

Sol : $\frac{a_1}{a_2}=\frac{b_1}{b_2}=\frac{c_1}{c_2}$

$$\frac{2}{4}=\frac{-1}{-2}=\frac{-3}{5k}\Rightarrow k=\frac{-6}{5}\Rightarrow5k^2+6=\frac{66}{5}$$

W.E-14 : *If the planes $x+2y+kz=0$ and $2x+y-2z=0$ are at right angles, then the value of k is*

Sol : $a_1a_2+b_1b_2+c_1c_2=0$

$1(2)+2(1)+k(-2)=0$

$\Rightarrow4-2k=0$, $\Rightarrow k=2$

Equations of planes bisecting the angles between given planes:

➢ i) Equations of two planes bisecting the angles between the planes

$a_1x+b_1y+c_1z+d_1=0$ and

$a_2x+b_2y+c_2z+d_2=0$ are

$$\frac{a_1x+b_1y+c_1z+d_1}{\sqrt{a_1^2+b_1^2+c_1^2}}=\pm\frac{a_2x+b_2y+c_2z+d_2}{\sqrt{a_2^2+b_2^2+c_2^2}}$$

ii) If $d_1,d_2>0$

Condition	Acute	Obtuse
$a_1a_2+b_1b_2+c_1c_2>0$	$-$	$+$
$a_1a_2+b_1b_2+c_1c_2<0$	$+$	$-$

iii) a) The Bisector planes are perpendicular to each other

b) Positive sign bisector is the bisector containing the origin.

The projection of line segment on a line (plane):

➢ Let P, Q be two points and L be a line (Π plane). If M, N are feet of perpendiculars from P,Q to the line L (to the plane Π) respectively then MN is called projection of PQ on the line L(the plane Π). The length of projection of PQ is always non-negative.

Some standard results:

➢ i) If $P_1=a_1x+b_1y+c_1z+d_1=0$ and $P_2=a_2x+b_2y+c_2z+d_2=0$ are two intersecting planes then the plane passing through their line of intersection is $P_1+kP_2=0$ where k is any constant.

ii) The equation of plane which bisects the join of the points (x_1,y_1,z_1) and (x_2,y_2,z_2) at right angles is

$$\sum(x_1-x_2)\left\{x-\frac{1}{2}(x_1+x_2)\right\}=0$$

iii) If a plane meets the coordinate axes in A,B,C such that the centroid of the triangle ABC is the point (p,q,r) then the equation of the plane is

$$\frac{x}{p}+\frac{y}{q}+\frac{z}{r}=3$$

iv) Two systems of rectangular axes have the same origin.If a plane cuts them at distance a, b, c and a_1,b_1,c_1 respectively from the origin, then

$$a^{-2}+b^{-2}+c^{-2}=a_1^{-2}+b_1^{-2}+c_1^{-2}$$

v) A variable plane is at a constant distance 'p' from the origin and meets the axes in A, B and C. The locus of the centroid of the triangle ABC is $x^{-2}+y^{-2}+z^{-2}=9p^{-2}$

vi) A variable plane is at a constant distance 'p' from the origin and meets the axes in A, B and C. The locus of the centroid of the tetrahedron OABC is $x^{-2}+y^{-2}+z^{-2}=16p^{-2}$

vii) A variable plane passes through a fixed point (α,β,γ) and meets the coordinate axes in A, B, C. Then the locus of the point of intersection of the planes through A, B, C parallel to the coordinate planes is

$$\frac{\alpha}{x}+\frac{\beta}{y}+\frac{\gamma}{z}=1$$

viii) A variable plane at a constant distance 'p' from the origin meets the axes in A, B and C. Through A, B, C planes are drawn parallel to the coordinate planes.Then the locus of their point of intersection is $x^{-2}+y^{-2}+z^{-2}=p^{-2}$

ix) A point P moves on the fixed plane $\frac{x}{a}+\frac{y}{b}+\frac{z}{c}=1$. The plane through P, perpendicular to OP meets the coordinate axes in A, B and C.Then the locus of the point of intersection of the planes through A, B, C parallel to the coordinate planes is

$$\frac{1}{x^2}+\frac{1}{y^2}+\frac{1}{z^2}=\frac{1}{ax}+\frac{1}{by}+\frac{1}{cz}.$$

x) The planes $x=\pm a, y=\pm b$ and $z=\pm c$ form a rectangular parallelopiped.

xi) A parallelopiped is formed by the planes drawn through the points (x_1,y_1,z_1) and (x_2,y_2,z_2) parallel to the co-ordinate planes. The length of a diagonal of the parallelopiped $=\sqrt{a^2+b^2+c^2}$

Here $a=x_2-x_1, b=y_2-y_1, c=z_2-z_1$

LEVEL - I

Cartesian equation of a plane:

1. The plane which passes through the point (-1, 0, -6) and perpendicular to the line whose d.r's are(6, 20, -1) also passes through the point

1) (1, 1, -26) 2) (0, 0, 0)
3) (2, 1, -32) 4) (1, 1, 1)

2. Equation of the plane through the mid-point of the join of A(4,5,-10) and B(-1,2,1) and perpendicular to AB is

1) $5x+3y-11z+\frac{135}{2}=0$

2) $5x+3y-11z=\frac{135}{2}$

3) $5x+3y+11z=135$

4) $5x+3y-11z+\frac{185}{2}=0$

3. In the space the equation $by+cz+d=0$ represents a plane perpendicular to the plane

1) YOZ 2) ZOX 3) XOY 4) Z = k

4. The equation of the plane passing through a point with position vector $\hat{i}+2\hat{j}+3\hat{k}$ and parallel to the plane $\vec{r}.\left(3\hat{i}+4\hat{j}+5\hat{k}\right)=0$ is

1) $3x+4y-5z+26=0$
2) $3x+4y+5z-26=0$
3) $3x-4y+5z-26=0$
4) $3x+4y-5z-26=0$

5. Distance between two parallel planes $2x+y+2z=8$ and $4x+2y+4z+5=0$ is **(JEE MAINS 2013)**

1) $\frac{3}{2}$ 2) $\frac{9}{2}$ 3) $\frac{7}{2}$ 4) $\frac{5}{2}$

Foot and image:

6. If the foot of perpendicular from (0, 0, 0) to a plane is (1, 2, 2) then the equation of the plane is

1) $-x+2y+8z-9=0$

2) $x+2y+2z-9=0$

3) $x+y+z-5=0$ 4) $x+2y=3z+1=0$

7. The foot of the perpendicular from (7, 14, 5) to $2x+4y-z=2$ is

1) $(-1,1,0)$ 2) $(1,2,8)$

3) $(2,-1,-2)$ 4) $(1,2,3)$

8. The image of the point $(-1,3,4)$ in the plane $x-2y=0$ is

1) $\left(-\frac{17}{3},-\frac{19}{3},1\right)$ 2) $\left(\frac{9}{5},-\frac{13}{5},4\right)$

3) $\left(-\frac{17}{3},-\frac{19}{3},4\right)$ 4) $(15,11,4)$

Ratio formula:

9. If the points (1,-1,1) and (-2,0,5) with respect to the plane $2x+3y-z+7=0$ lie on

1) opposite sides 2) same side
3) on the plane 4) in side the plane

10. The ratio in which the plane $\bar{r}.(\bar{i}-2\bar{j}+3\bar{k})=17$ divides the line joining the points $-2\hat{i}+4\hat{j}+7\hat{k}$ and $3\hat{i}-5\hat{j}+8\hat{k}$ is

1) 1 : 5 2) 1 : 10 3) 3 : 5 4) 3 : 10

Normal form of a plane:

11. For the plane $\Pi \equiv 2x+3y+5z+10=0$, the point $(2,3,-5)$ lie in the

1) Opposite to the origin side
2) Origin side 3) Plane 4) can not say

12. The normal form of $2x-2y+z=5$ is

1) $12x-4y+3z=39$

2) $-\frac{6}{7}x+\frac{2}{7}y+\frac{3}{7}z=1$

3) $\frac{12}{13}x-\frac{4}{13}y+\frac{3}{13}z=3$

4) $\frac{2}{3}x-\frac{2}{3}y+\frac{1}{3}z=\frac{5}{3}$

Perpendicular distance:

13. A plane passes through $(2,3-1)$ and is perpendicular to the line having dr's $(3,-4,7)$. The perpendicular distance from the origin to this plane is **(EAM-2011)**

1) $\frac{3}{\sqrt{74}}$ 2) $\frac{5}{\sqrt{74}}$ 3) $\frac{6}{\sqrt{74}}$ 4) $\frac{13}{\sqrt{74}}$

14. An equation of a plane parallel to the plane $x-2y+2z-5=0$ and at a unit distance from the origin is **(AIE-2012)**

1) $x-2y+2z-7=0$ 2) $x-2y+2z+5=0$

3) $x-2y+2z-3=0$ 4) $x-2y+2z+1=0$

Intercept form of a plane:

15. The equation of a plane which cuts equal intercepts of unit length on the axes is

1) $x+y+z=0$ 2) $x+y+z=1$

3) $2x+y-z=1$ 4) $\frac{x}{a}+\frac{y}{a}+\frac{z}{a}=1$

16. If the plane $7x+11y+13z=3003$ meets the coordinate axes in A, B, C then the centroid of the ΔABC is

1) $(143,91,77)$ 2) $(143,77,91)$

3) $(91,143,77)$ 4) $(143,66,91)$

17. If the areas of triangles formed by a plane with the positive $X,Y:Y,Z:Z,X$ axes respectively are 12, 9, 6 sq. unit respectively then the equation of the plane is

1) $\frac{x}{4}+\frac{y}{6}+\frac{z}{3}=1$ 2) $\frac{x}{6}+\frac{y}{3}+\frac{z}{4}=1$

3) $\frac{x}{3}+\frac{y}{4}+\frac{z}{6}=1$ 4) $\frac{x}{3}+\frac{y}{6}+\frac{z}{4}=1$

Angle between two planes:

18. **If the planes $x+2y+kz=0$ and 2x+y-2z+3=0 are at right angles, then the values of k is**

1) $-\frac{1}{2}$ 2) $\frac{1}{2}$ 3) -2 4) 2

19. **The angle between the planes $2x+y+z=3, x-y+2z=5$ is**

1) $\frac{\pi}{2}$ 2) $\frac{\pi}{6}$ 3) $\frac{3\pi}{4}$ 4) $\frac{\pi}{3}$

20. **If the planes $2x+3y-z+5=0$, $x+2y-kz+7=0$ are perpendicular then $k=$**

1) 4 2) 6 3) 8 4) -8

LEVEL-I -KEY

1) 2	2) 2	3) 1	4)2
5) 3	6) 2	7) 2	8) 2
9) 1	10)4	11) 1	12) 4
13) 4	14) 3	15) 2	16) 1
17) 1	18) 4	19) 4	20) 4

LEVEL-II

Cartesian equation of a plane:

1. **The product of the d.r's of a line perpendicular to the plane passing through the points (4,0,0), (0,2,0) and (1,0,1) is**

1) 6 2) 2 3) 0 4) 1

2. **A plane which passes through the point (3, 2, 0) and the line $\frac{x-4}{1}=\frac{y-7}{5}=\frac{z-4}{4}$ is**

1) $x-y+z=1$ 2) $x+y+z=5$

3) $x-2y-z=1$ 4) $2x-y+z=5$

3. **The equation of the plane parallel to the plane $2x+3y+4z+5=0$ and passing through the point (1,1,1) is**

1) $2x+3y+4z-9=0$

2) $2x+3y+4z+9=0$

3) $2x+3y+4z+7=0$

4) $2x+3y+4z-7=0$

4. **The equation of the plane passing through the point (3, -6, 9) and perpendicular to the x-axis is**

1) $x+2=0$ 2) $y-3=0$

3) $z-7=0$ 4) $x-3=0$

5. **Distance between two parallel planes $7x+4y-4z+3=0$ and $14x+8y-8z-12=0$ is**

1) $\frac{15}{9}$ 2) 1 3) $\frac{9}{15}$ 4) $\frac{1}{2}$

Foot and image:

6. **If the foot of the perpendicular from $(0,0,0)$ to a plane is $(1,2,3)$, then the equation of the plane is (EAM-2012)**

1) $2x+y+3z=14$ 2) $x+2y+3z=14$

3) $x+2y+3z+14=0$ 4) $x+2y-3z=14$

7. **The image of the point $(3,2,1)$ in the plane $2x-y+3z=7$ is (EAM-2009)**

1) $(1,2,3)$ 2) $(2,3,1)$ 3) $(3,2,1)$ 4) $(2,1,3)$

8. **If $(2,3,-1)$ is the foot of the perpendicular from $(4,2,1)$ to a plane, the equation of the plane is**

1) $2x-y-2z-3=0$ 2) $2x+y-2z-9=0$

3) $2x+y+2z-5=0$ 4) $2x-y+2z+1=0$

9. **If the points $(1,1,p)$ and $(-3,0,1)$ be equidistant from the plane $\bar{r}.(3\bar{i}+4\bar{j}+12\bar{k})+13=0$ then the value of P =**

1) $-\frac{1}{3}$ 2) 6 3) 3 4) $\frac{1}{3}$

Ratio formula:

10. **The ratio in which the line joining $(2,-4,3)$ and $(-4,5,-6)$ is divided by the plane $3x+2y+z-4=0$ is (EAM-2011)**

1) 2 : 1 2) 4 : 3 3) -1 : 4 4) 2 : 3

11. **For the plane $\Pi \equiv 4x-3y+2z-3=0$, the points $A(-2,1,2), B=(3,1,-2)$**

1) Lie on the same side of $\Pi=0$

2) Lie on the opposite sides of $\Pi=0$

3) Lie on the normal to $\Pi=0$ 4) Lie on $\Pi=0$

Normal form of a plane:

12. The d.c's of the normal to the plane $2x-y+2z+5=0$ are

1) (3, -2, 6) 2) $\left(\frac{2}{7},\frac{3}{7},\frac{-6}{7}\right)$

3) $\left(\frac{3}{7},\frac{-2}{7},\frac{6}{7}\right)$ 4) $\left(\frac{2}{3},\frac{-1}{3},\frac{2}{3}\right)$

Perpendicular distance:

13. The perpendicular distance from the origin to the plane $x-2y+2z-9=0$ is

1) 3 2) 6 3) 4 4) 2

14. The length of the perpendicular from the point $(7,14,5)$ to the plane $2x+4y-z=2$ is

1) $21\sqrt{3}$ 2) $3\sqrt{21}$ 3) $\sqrt{321}$ 4) $5\sqrt{2}$

Intercept form of a plane:

15. 5, 7, are the intercepts of a plane on the Y - axis, Z - axis respectively, if the plane is parallel to the X- axis then the equation of that plane is

1) $5y+7z=35$ 2) $7y+5z=1$

3) $\frac{y}{5}+\frac{z}{7}=35$ 4) $7y+5z=35$

16. The intercepts of the plane $2x-3y+5z-30=0$ are

1) $15,-10,6$ 2) $5,10,6$

3) $\frac{1}{8},\frac{-1}{6},\frac{1}{4}$ 4) $3,-4,6$

17. The area of the triangle formed by $\frac{x}{4}+\frac{y}{3}-\frac{z}{2}=1$ with X-axis and Y-axis is

1) 2 2) 3 3) 6 4) 12

Angle between two planes:

18. If the planes $2x+3y+4z+7=0$ and $4x+ky+8z+1=0$ are parallel then $k=$

1) 2 2) 4 3) 5 4) 6

19. The planes $2x-y+4z=5$ and $5x-(2.5)y+10z=6$ are

1) Perpenducular 2) parallel

3) intersect Y-axis 4) passes through (0,0,5/4)

20. If $\lambda x+4y+5z=7$, $4x+4\lambda y+10z-14=0$ represent the same plane then the value of $\lambda=$

1) 1 2) 2 3) 0 4) 3

LEVEL-II-KEY

1) 1	2) 1	3) 1	4) 4	5) 2
6) 2	7) 3	8) 4	9)1	10) 3
11) 2	12) 4	13)1	14) 2	15) 4
16) 1	17)3	18) 4	19) 2	20) 2

LEVEL-III

Cartesian equation of a plane:

1. If the equation of the plane passing through the points (1,2,3), (-1,2,0) and perpendicular to the ZX - plane is $ax+by+cz+d=0$ $(a>0)$ then

1) $a=0\ and\ c=0$ 2) $a+d=0$

3) $c+d-5=0$ 4) $a+c+d-4=0$

2. The equation of the plane through the line of intersection of planes $ax+by+cz+d=0$, $a'x+b'y+c'z+d'=0$ and parallel to the lines y = 0 = z is

1) $(ab'-a'b)x+(bc'-b'c)y+(ad'-a'd)=0$

2) $(ab'-a'b)y+(ac'-a'c)z+(ad'-a'd)=0$

3) $(ab'-a'b)x+(bc'-b'c)z+(ad'-a'd)=0$

4) $(ab'-a'b)x-(bc'-b'c)y+(ad'+a'd)=0$

3. A plane π passes through the point (1,1,1). If b,c, a are the dr's of a normal to the plane, where a, b, c $(a<b<c)$ are the prime factors of 2001, then the equation of the plane π is

1) $29x+31y+3z=63$

2) $23x+29y-29z=23$

3) $23x+29y+3z=55$

4) $31x+27y+3z=71$

4. The dr's of a normal to the plane through (1,0,0), (0,1,0) which makes an angle of $\frac{\pi}{4}$ with the plane $x+y=3$ are

1) $1,\sqrt{2},1$ 2) $1,1,\sqrt{2}$ 3) $1,1,2$ 4) $\sqrt{2},1,1$

5. Let $A(1,1,1), B(2,3,5)\ and\ C(-1,0,2)$ be three points, then equation of a plane parallel to the plane ABC which is at a distance 2 units is

1) $2x-3y+z+2\sqrt{14}=0$

2) $2x-3y+z-\sqrt{14}=0$

3) $2x-3y+z+2=0$

4) $2x-3y+z-2=0$

Intercept form of a plane:

6. **A variable plane intersects the coordinate axes at A,B,C and is at a constant distance 'p' from 0(0,0,0). Then the locus of the centroid of the tetrahedron OABC is**

1) $\frac{1}{x^2}+\frac{1}{y^2}+\frac{1}{z^2}=\frac{1}{p^2}$ 2) $\frac{1}{x^2}+\frac{1}{y^2}+\frac{1}{z^2}=\frac{4}{p^2}$

3) $\frac{1}{x^2}+\frac{1}{y^2}+\frac{1}{z^2}=\frac{16}{p^2}$ 4) $\frac{1}{x^2}+\frac{1}{y^2}+\frac{1}{z^2}=16p^2$

7. **The equation to the plane through the line of intersection of $2x+y+3z-2=0$, $x-y+z+4=0$ such that each plane is at a distance of 2 unit from the origin is**

1) $x+y+2z+13=0, x+y+z-3=0$

2) $2x+y-2z+3=0, x-2y-2z-3=0$

3) $15x-12y+16z+50=0$, $x+2y+2z-6=0$

4) x-y+2z-13=0, x+y-z-3=0

8. **The equation of the plane which is parallel to X-axis and making intercepts 3 and 8 on Y and Z-axes respectively is**

1) $3y+8z=24$ 2) $3y-8z=24$

3) $8y-3z=24$ 4) $8y+3z=24$

9. **The sum of the intercepts of the plane which bisects the line segment joining (0,1,2) and (2,3,0) perpendicularly is**

1) 2 2) 4 3) 6 4) 12

10. **A plane meets the coordinate axes at A, B, C so that the centroid of the triangle ABC is (1, 2, 4). Then the equation of the plane is (EAM-2010)**

1) $x+2y+4z=12$ 2) $4x+2y+z=12$

3) $x+2y+4z=3$ 4) $4x+2y+z=3$

11. **The reflection of the plane $2x-3y+4z-3=0$ in the plane $x-y+z-3=0$ is the plane**

1) $4x-3y+2z-15=0$

2) $x-3y+2z-15=0$

3) $4x+3y-2z+15=0$

4) $4x+3y+2z+15=0$

LEVEL-III-KEY

1) 4 2) 2 3) 3 4) 2 5) 1 6) 3
7) 3 8) 4 9) 1 10) 2 11) 1

LEVEL-IV

Cartesian equation of a plane:

1. **The d.r's of the line of intersection of the planes $x+y+z-1=0$ and $2x+3y+4z-7=0$ are**

1) 1, 2, -3 2) 2, 1, -3

3) 4, 2, -6 4) 1, -2, 1

2. **The vertices of a tetrahedron are A(3,4,2) B(1,2,1) C(4,1,3) D(-1,-1,3). The height of A above the base BCD.**

1) $\frac{27}{\sqrt{237}}$ 2) $\frac{23}{\sqrt{237}}$ 3) $\frac{20}{\sqrt{237}}$ 4) $\frac{27}{\sqrt{247}}$

3. **The equation of the plane which passes through the line of intersection of the planes $2x-y=0$ and $3z-y=0$ and is perpendicular to the plane** $4x+5y-3z=8$ is

1) $28x-17y+9z=0$ 2) $28x+17y+9z=0$

3) $2x+17y-9z=0$ 4) $2x-y-z=0$

4. **The dr's of a normal to the plane passing through $(0,0,1),(0,1,2)$ and $(1,2,3)$ are (EAM-2002)**

1) $(0,1,-1)$ 2) $(1,0,-1)$

3) $(0,0,-1)$ 4) $(1,0,0)$

5. **The plane $2x+3y+kz-7=0$ is parallel to the line whose direction ratios are $(2,-3,1)$ then** $k=$

1) 5 2) 8 3) 1 4) 0

Intercept form of a plane:

6. **A variable plane is at a constant distance 3p from the origin and meets the axes in A, B**

and C. The locus of the centroid of the triangle ABC is

1) $x^{-2}+y^{-2}+z^{-2}=p^{-2}$
2) $x^{-2}+y^{-2}+z^{-2}=4p^{-2}$
3) $x^{-2}+y^{-2}+z^{-2}=16p^{-2}$
4) $x^{-2}+y^{-2}+z^{-2}=9p^{-2}$

7. **The equation of the plane through the line of intersection of the planes** $x-2y+3z-1=0$, $2x+y+z-2=0$ **and the point** $(1,2,3)$ **is**

1) $7x-9y+8z=0$ 2) $7x+y+8z=0$
3) $x+3y-2z-1=0$ 4) x-3y-2z+1=0

8. **The equation of the plane which is parallel to Y-axis and making intercepts of lengths 3 and 4 on X-axis and Z-axis is**

1) $2x+2z=20$ 2) $4x+3z=12$
3) $4x-3z=12$ 4) $6x+13z=15$

9. **If 5, 7, 6 are the sums of the X, Y intercepts; Y, Z intercepts, Z, X intercepts respectively of a plane then the perpendicular distance from the origin to that plane is**

1) $\frac{144}{61}$ 2) $\frac{12}{\sqrt{61}}$ 3) $\frac{\sqrt{61}}{12}$ 4) $\frac{61}{144}$

10. **The equation of the plane through the line of intersection of the planes** $2x+3y+4z-7=0$, $x+y+z-1=0$ **and perpendicular to the plane** $x-5y+3z-2=0$ **is**

1) $7x-y-6z-17=0$
2) $x-y-6z-27=0$
3) $x+2y+3z-6=0$
4) x+y+6z-27=0

11. **The equations of bisectors of angles between YZ-plane and XZ-plane is**

1) $x-z=0, x+2z=0$
2) $x-z+2=0$
3) $x+z=0, x-z=0$
4) $x+y=0, x-y=0$

LEVEL-IV-KEY

1) 4 2) 2 3) 1 4) 1 5) 1 6) 1
7) 3 8) 2 9) 2 10) 3 11) 4

LEVEL-V

1. **If the angles made by the normal of the plane** $2x+3y-4z-16=0$ **with the coordinates axes X, Y, Z are** $Cos^{-1}k_1, Cos^{-1}k_2, Cos^{-1}k_3$, **then** k_1,k_2,k_3 **respecitvely are**

1) $\frac{2}{\sqrt{29}},\frac{3}{\sqrt{29}},\frac{4}{\sqrt{29}}$ 2) $\frac{2}{\sqrt{29}},-\frac{3}{\sqrt{29}},\frac{4}{\sqrt{29}}$
3) $\frac{2}{\sqrt{29}},\frac{3}{\sqrt{29}},-\frac{4}{\sqrt{29}}$ 4) $\frac{1}{2},\frac{1}{3},-\frac{1}{4}$

2. **The two planes represented by** $12x^2-2y^2-6z^2-7yz+6zx-2xy=0$ **are**

1) $2x+y+2z=0, 6x-2y+3z=0$
2) $2x-y+2z=0, 6x+2y-3z=0$

LEVEL - II (C.W)

3) $2x-y+2z+4=0, 6x+2y-3z=0$
4) $2x-y+2z=0, 6x+2y-3z+1=0$

3. **If the plane** $4(x-1)+k(y-2)+8(z-5)=0$ **contains the line** $\frac{x-1}{2}=\frac{y-2}{4}=\frac{z-5}{3}$, **then** k **is**

1) 2 2) 4 3) -8 4) 8

4. **If the plane** $3(x-2)+(y-2)+6(z+3)=0$ **contains the line** $\frac{x-2}{a}=\frac{y-2}{b}=\frac{z+3}{1}$ **whose inclination with X-axis is** 60^0, **then it satisfies the equation**

1) $6a^2+36a+37=0$
2) $36a^2+37=0$

3) $36a^2+37a+36=0$

4) $a+3=0$

5. **The angle between the planes represented by $2x^2-6y^2-12z^2+18yz+2zx+xy=0$ is**

1) $Cos^{-1}\left(\frac{16}{21}\right)$ 2) $Cos^{-1}\left(\frac{17}{21}\right)$

3) $Cos^{-1}\left(\frac{19}{21}\right)$ 4) $\frac{\pi}{2}$

6. **If the equation of the plane passing through the line of intersection of the planes $ax+by+cz+d=0, a_1x+b_1y+c_1z+d_1=0$ and perpendicular to the XY-plane is $px+qy+rz+s=0$ then S=**

1) dc_1-d_1c 2) dc_1+d_1c

3) dd_1+cc_1 4) $aa_1+bb_1+cc_1$.

7. **If the points (1, 1, -3) and (1, 0, -3) lie on opposite sides of the plane $x+y+3z+d=0$ then**

1) $d<7$ 2) $d>8$

3) $7<d<8$ 4) $d<7$ or $d>8$.

8. **P is a point such that the sum of the squares of its distances from the planes $x+y+z=0$, $x+y-2z=0, x-y=0$ is 5 then the locus of P is**

1) $x^2+y^2+z^2=10$ 2) $x^2+y^2+z^2=25$

3) $x^2+y^2+z^2=5$ 4) $x^2+y^2+z^2=50$

9. **The areas of triangles formed by a plane with the positive $X,Y;Y,Z;Z,X$ axes respectively are 12, 9, 6 square units then the equation of the plane is**

1) $\frac{x}{4}+\frac{y}{6}+\frac{z}{3}=1$ 2) $\frac{x}{6}+\frac{y}{3}+\frac{z}{4}=1$

3) $\frac{x}{4}+\frac{y}{4}+\frac{z}{6}=1$ 4) $\frac{x}{3}+\frac{y}{6}+\frac{z}{4}=1$

10. **The plane $ax+by+cz+(-3)=0$ meet the co-ordinate axes in A,B,C. Then centroid of the triangle is**

1) (3a,3b,3c) 2) $\left(\frac{3}{a},\frac{3}{b},\frac{3}{c}\right)$

3) $\left(\frac{a}{3},\frac{b}{3},\frac{c}{3}\right)$ 4) $\left(\frac{1}{a},\frac{1}{b},\frac{1}{c}\right)$

11. Equation of the plane passing through the point (-1,3,2) and perpendicular to each of the planes $x+2y+3z=5$ and $3x+3y+z=0$ is

1) $7x+8y-3z=0$

2) $7x-8y-3z=-37$

3) $7x-8y+3z+25=0$

4) $7x+8y+3z=23$

12. **If P= (0,1,0) and Q = (0,0,1) then the projection of PQ on the plane $x+y+z=3$ is**

1) 2 2) $\sqrt{2}$ 3) 3 4) $\sqrt{3}$

13. **A parallelopiped is formed by the planes drawn through the points (2, 3, 5) and (5, 9, 7) parallel to the coordinate planes. The length of diagonal of the paralleopiped is**

1) 7 2) $\sqrt{38}$ 3) $\sqrt{155}$ 4) $\sqrt{7}$

LEVEL-V - KEY

1) 3 2) 2 3) 3 4) 1 5) 1 6) 1 7) 3
8) 3 9) 1 10) 4 11) 3 12) 2 13) 1

LEVEL-VI

1. **Match the following**

List-I	List-II
a) The image of (2, 1, 4) in the plane $2x-y+z+5=0$	i) $(-2,3,2)$
b) The foot of perpendicular from (2, 1, 4) in the plane $2x-y+z+5=0$	ii) $(-4,1,4)$
c) The point lying on $2x-y+z+5=0$	iii) $(-6,5,0)$

Which of the following is correct?

1) a-iii, b-ii, c-i 2) a-i, b-iii, c-ii

3) a-i, b-ii, c-iii 4) a-iii, b-i, c-ii

2. **Observe the following Statements:**

Statement I : The image of (0, 0, 0) in the plane $2x-3y+6z-49=0$ is (4, -6, 12)

Statement II : The foot of the perpendicular from (1, 2, 3) to the plane $x+y-2z-9=0$ is (3, 4, -1)

Which of the following is correct?

1) I is ture, II is true 2) I is true, II is false

3) Both I and II are false 4) I is false, II is true

3. α,β,γ **are the angles made by the normal of the plane** $2x-3y+6z-6=0$ **with axes X,Y,Z respectively. Match the cosine of the angles given in List I with corresponding values given in List II.**

List-I	List-II
a) $\cos\alpha$	i) $6/7$
b) $\cos\beta$	ii) $-3/7$
c) $\cos\gamma$	iii) $2/7$

The correct match from list I to list II is

1) a-i, b-ii, c-iii 2) a-ii, b-i, c-iii
3) a-ii, b-iii, c-i 4) a-iii, b-ii, c-i

4. **The coordinates of a point P are (1, 2, -1). Match the plane given in List I with corresponding perpendicular distance from the point P to their planes given in List II**

List-I	List-II
a) $2x+y-z+2=0$	i) 1
b) $x-y+6z=0$	ii) $\frac{7}{\sqrt{6}}$
c) $3x+2y+6z+6=0$	iii) $\frac{7}{\sqrt{38}}$

Which of the following is correct?

1) a-ii, b-i, c-iii 2) a-i, b-ii, c-iii
3) a-iii, b-ii, c-i 4) a-ii, b-iii, c-i

ASSERTION & REASON

The following questions consist of two statements one labelled as , Assertion (A) and the other labelled as Reason (R) Y ou are to examine these two statements carefully and decide if the Assertion (A) and the Reason (R) are individually true and if so,whether the Reason (R) is the correct explanation for the given Assertion (A) select your answer to these items using the codes given below and then select the correct option

Codes:

(A) *Both A and R are individually true and R is the correct explanation of A*

(B) *Both A and R individually true but R is not the correct explanation of A*

(C) *A is true but R is false*

(D) *A is false but R is true*

5. **Assertion (A)** ***:*** **The equation of the plane through the intersection of the planes** $x+y+z=6$ **and** $2x+3y+4z+5=0$ ***a*****nd the point (4,4,4) is** $29x+23y+17z=276$

Reason (R) : Equation of the plane through the line of intersection of the planes $P_1=0$ **and** $P_2=0$ **is** $P_1+\lambda P_2=0$ **(**$\lambda\neq0$**)**

1) A 2) B 3) C 4) D

6. **Assertion (A) : The points** $(2,1,5)$ **and** $(3,4,3)$ **lie on opposite side of the plane** $2x+2y-2z-1=0$

Reason (R) : The algebraic perpendicular distance from the given points to the line have opposite sign

1) A 2) B 3) C 4) D

Linked comprehensive type passage:

Let two planes $p_1: 2x-y+z=2$, and $p_2: x+2y-z=3$ are given .

7. **The equation of the plane passing through the intersection of** P_1 **and** P_2 **and the point** $(3,2,1)$ **is**

1) $3x-y+2z-9=0$
2) $x-3y+2z+1=0$
3) $2x-3y+z-1=0$
4) $4x-3y+2z-8=0$

8. **Equation of the plane which passes through the point** $(-1,3,2)$ **and is perpendicular to each of the planes** P_1 **and** P_2 **is**

1) $x+3y-5z+2=0$
2) $x+3y+5z-18=0$
3) $x-3y-5z+20=0$
4) $x-3y+5z=0$

9. **The equation of the acute angle bisector of planes** P_1 **and** P_2 **is**

1) $x-3y+2z+1=0$
2) $3x+y-5=0$
3) $x+3y-2z+1=0$
4) $3x+z+7=0$

10. **The equation of the bisector of angle of the**

planes P_1 and P_2 which not containing origin is

1) $x-3y+2z+1=0$ 2) $x+3y=5$

3) $x+3y+2z+2=0$ 4) $3x+y=5$

11. The image of plane P_1 in the plane mirror P_2 is

1) $x+7y-4z+5=0$

2) $3x+4y-5z+9=0$

3) $7x-y+2z-9=0$

4) $7x+y+9z+9=0$

LEVEL-VI-KEY

1) 4	2) 1	3) 4	4) 4
5) 1	6) 1	7) 2	8) 3
9) 1	10) 4	11) 3	

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